Introduction to Cloud Computing

Note:

1. The following contents are indicative and just for quick reference.
2. You shall write elaborately in the exam with proper examples/contexts/scenarios.
3. Prepare a business context / scenario for every concept.

MODULE 4

1. Cloud Security: The shared responsibility model

The cloud shared responsibility model defines security duties between cloud providers (CSPs) and customers, splitting them into "Security *of* the Cloud" (provider's job, like physical infrastructure) and "Security *in* the Cloud" (customer's job, like data, applications, and configurations). Responsibilities shift based on the service model (IaaS, PaaS, SaaS), with customers owning more as services become more managed, ensuring a crucial framework to prevent security gaps and misconfigurations.

Core Principles

- **Provider (CSP) Responsibility:** Securing the underlying cloud infrastructure (hardware, virtualization, physical facilities).
- **Customer Responsibility:** Securing their own data, applications, identity/access, endpoints, and configurations *within* the cloud.

By Cloud Service Model (IaaS, PaaS, SaaS)

- **Infrastructure as a Service (IaaS):** CSP secures hardware/virtualization; customer secures OS, middleware, apps, data, access.
- **Platform as a Service (PaaS):** CSP secures infrastructure & platform; customer secures apps, data, access, and platform configurations.
- **Software as a Service (SaaS):** CSP manages most; customer secures data, user access, and endpoint devices.

Why it Matters

- **Defines Accountability:** Clearly assigns who handles security tasks.
- **Prevents Gaps:** Helps avoid security holes from misunderstandings.
- **Mitigates Breaches:** Misconfigurations are a top cause of breaches; understanding this model prevents them.

Example (AWS services):

AWS: Security OF the Cloud

AWS protects the infrastructure running all services:

- **Physical:** Data centers, hardware, networking equipment
- **Global infrastructure:** Regions, Availability Zones, Edge locations

- **Core services:** Compute, storage, database, networking
- **Software:** Virtualization and underlying software

You: Security IN the Cloud

You are responsible for everything you put in AWS:

- **Your data:** What you store and how you encrypt it
- **Access control:** Who can access your resources
- **Configuration:** Operating system, network, and firewall settings
- **Applications:** The software you run and how you secure it

Fig. Responsibilities of the user and provider

Layer	Infrastructure as a Service (IaaS)	Platform as a Service (PaaS)	Software as a Service (SaaS)
Data	Customer responsibility	Customer responsibility	Customer responsibility
Application	Customer responsibility	Customer responsibility	Cloud provider responsibility
Operating system	Customer responsibility	Cloud provider responsibility	Cloud provider responsibility
Virtualization	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility
Servers	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility
Storage	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility
Network	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility
Physical	Cloud provider responsibility	Cloud provider responsibility	Cloud provider responsibility

Cloud Management Essentials: Security, Visibility, Performance, and Cost Control

As organizations increasingly migrate their infrastructure and applications to the cloud, effective cloud management becomes essential. Four critical areas play a major role in ensuring secure, efficient, and cost-effective cloud operations: Identity and Access Management (IAM), Monitoring and Logging, Tools for Tracking Resource Usage and Application Health, and Cost Management. Together, these components help organizations maintain control, visibility, and accountability in dynamic cloud environments.

2. Identity and Access Management (IAM)

Identity and Access Management (IAM) is a framework of policies, technologies, and processes that ensures the right individuals and systems have appropriate access to cloud resources. IAM focuses on authentication (verifying who a user is) and authorization (determining what they are allowed to do). In cloud environments, IAM is critical because resources are accessible over the internet and often shared across teams and regions. Cloud providers such as AWS, Microsoft Azure, and Google Cloud allow organizations to define users, roles, and permissions with fine-grained control.

Example Scenario:

A software company hosts its application on the cloud with separate teams for development, testing, and production support. Using IAM, developers are granted permission to deploy code only in the development environment, while operations staff have access to production servers. If a developer's account is compromised, IAM policies prevent unauthorized changes to critical production systems. By enforcing the principle of **least privilege**, IAM reduces security risks, prevents accidental misuse of resources, and supports compliance with regulatory requirements.

Features of IAM Users in Cloud: IAM users represent individual identities (people or services) that interact with cloud resources. Key features include:

1. **Unique Identity**
   - Each IAM user has a unique username and credentials.
   - Helps track actions performed by a specific user.
2. **Authentication Mechanisms**
   - Supports passwords, access keys, and multi-factor authentication (MFA).
   - Enhances security by requiring more than one verification method.
3. **Fine-Grained Access Control**
   - Permissions can be assigned at a very detailed level (service, resource, or action).
   - Prevents users from accessing unauthorized resources.
4. **Policy-Based Permissions**
   - Access is controlled using policies written in JSON or similar formats.
   - Policies define what actions a user can or cannot perform.
5. **Role and Group Association**
   - IAM users can be added to groups or assigned roles.
   - Simplifies permission management for multiple users.
6. **Access Key Management**

- Users can be issued access keys for programmatic access (APIs, CLI).
- Keys can be rotated or revoked for security.

7. **Audit and Activity Tracking**
    - User actions are logged and monitored.
    - Supports compliance and security investigations.
8. **Temporary and Permanent Access**
    - Users can have long-term credentials or temporary session-based access.
    - Reduces risk of credential misuse.
9. **Cross-Service Access**
    - IAM users can access multiple cloud services using a single identity.
    - Improves operational efficiency.

Applications of IAM Users in Cloud: IAM users are widely used across cloud environments for security and operational control.

1. **User Authentication**
    - Verifies the identity of employees, administrators, and developers.
    - Example: A developer logs into the cloud console using an IAM user account.
2. **Authorization and Permission Management**
    - Controls what actions users can perform on cloud resources.
    - Example: A tester can view logs but cannot delete servers.
3. **Secure Access to Cloud Resources**
    - Protects virtual machines, databases, and storage services.
    - Example: Only authorized users can access sensitive customer data.
4. **Application and Service Access**
    - Enables applications to interact securely with cloud APIs.
    - Example: An application uploads files to cloud storage using IAM credentials.
5. **Multi-User Environment Management**
    - Supports collaboration across teams and departments.
    - Example: Finance, IT, and DevOps teams have different access levels.
6. **Compliance and Auditing**
    - Helps meet regulatory and compliance requirements.
    - Example: Audit logs show who accessed financial records and when.
7. **Automated Operations**
    - Used in scripts, CI/CD pipelines, and automation tools.
    - Example: A deployment script uses an IAM user to launch resources.
8. **Account Isolation and Security**
    - Prevents unauthorized access by enforcing least privilege.
    - Example: Interns have read-only access to non-critical resources.

3. Monitoring and Logging:

Monitoring and logging are essential for maintaining the reliability, security, and performance of cloud systems. **Monitoring** involves tracking real-time metrics such as CPU usage, memory consumption, and network traffic, while **logging** records events and activities for later analysis. Monitoring helps detect issues early, while logs provide detailed context for troubleshooting and auditing. Together, they enable organizations to respond quickly to outages, security incidents, or performance degradation.

Example Scenario:

An e-commerce website experiences slow checkout times during a seasonal sale. Monitoring tools reveal that CPU usage on application servers has reached critical levels. At the same time, logs show repeated timeout errors from the database. This information allows engineers to scale resources and optimize database queries, restoring performance before customers abandon their carts. Monitoring and logging also play a vital role in security, helping detect suspicious activities such as repeated failed login attempts or unauthorized API calls.

- Logging refers to the process of recording information about the events that occur within a system or application.
- Error messages, user actions, and system activities.
- Log data is typically stored in text files or a central log repository and can be used for a variety of purposes: *troubleshooting, auditing, and compliance.*
- Monitoring refers to the process of tracking the performance and health of a system or application.
- Resource usage, network traffic, and error rates.
- Collected in real time and can be used to identify and diagnose issues, predict and prevent problems, and optimize performance.

Best Practices for Implementing Monitoring & Logging Services

1. Centralized Logging: Collect all log data in a central location for easy access and analysis.
2. Use a Standard Log Format: Use a common log format such as JSON to facilitate parsing and analysis.
3. Implement Log Rotation: Regularly archive and remove old log files to save disk space and improve performance.
4. Use Log Levels: Use log levels such as "debug", "info", "warning", and "error" to categorize log messages and make it easier to filter and analyze the data.
5. Monitor Critical Metrics: Monitor key metrics such as system resource usage, network traffic, and error rates to quickly identify and troubleshoot issues.
6. Implement Alerting: Set up alerts to notify you when certain conditions are met, such as a high error rate or low disk space.

7. Use a Monitoring Solution: Use a monitoring tool such as Prometheus, Grafana, or New Relic to collect and analyze data.
8. Use a Log Aggregation Tool: Use a log aggregation tool such as ELK (Elasticsearch, Logstash, Kibana) or Splunk to search, analyze, and visualize log data.
9. Implement Security: Ensure that logs and monitoring data are protected from unauthorized access by implementing proper security measures such as encryption and authentication.
10. Regularly Review and Improve: Regularly review your logging and monitoring practices and make improvements as necessary to ensure that you are effectively monitoring your systems and applications.

4. Tools for tracking resource usage, performance, and application health:

Cloud providers offer specialized tools to track resource usage, system performance, and application health. These tools provide dashboards, alerts, and analytics to help teams understand how their cloud infrastructure is behaving.

Common capabilities include:

- Tracking compute, storage, and network usage
- Measuring application response times and error rates
- Setting automated alerts when thresholds are exceeded

Example Scenario:

A media streaming service uses cloud monitoring tools to track application health across multiple regions. When an application instance becomes unresponsive, the tool automatically triggers an alert and redirects traffic to healthy instances. This proactive monitoring ensures uninterrupted service for users and reduces manual intervention. Third-party tools can also integrate with cloud platforms to provide deeper insights, especially for complex, multi-cloud environments.

AWS Native Tools (discuss the following in your own words)

Amazon CloudWatch: Core monitoring service for metrics, logs, events, alarms, and custom dashboards.

AWS X-Ray: Traces requests as they travel through your application to pinpoint performance bottlenecks and analyze dependencies.

AWS CloudTrail: Logs all API calls and account activity for security, governance, and compliance auditing.

AWS Config: Tracks resource configuration changes and compliance.

VPC Flow Logs: Captures IP traffic info for your Virtual Private Cloud (VPC)

The primary AWS-native tools for logging and monitoring are Amazon CloudWatch and AWS CloudTrail, which provide performance data and API activity logs, respectively. Other AWS services and third-party tools complement these core functions.

Core AWS Logging and Monitoring Tools

- **Amazon CloudWatch**: This is the main service for monitoring the performance and health of your AWS resources and applications in real-time. It collects and tracks metrics, collects and monitors log files, and provides a unified view through dashboards and alarms.
    - **CloudWatch Logs**: Centralizes logs from all your systems, applications, and various AWS services (like EC2 instances, Lambda, and Route 53) for monitoring, searching, and secure archiving. It offers a powerful query language, CloudWatch Logs Insights, for analysis and a Live Tail feature for real-time troubleshooting.
    - **CloudWatch Metrics**: Collects time-ordered data points for performance analysis (e.g., CPU utilization, network traffic).
    - **CloudWatch Alarms**: Triggers notifications or automated actions (e.g., stopping an instance, sending an Amazon SNS notification) when a specific metric crosses a predefined threshold.
    - **CloudWatch Events (Amazon EventBridge)**: Delivers a near real-time stream of system events describing changes in AWS resources, allowing you to trigger automated responses.
- **AWS CloudTrail**: This service provides governance, compliance, and operational auditing of your AWS account by recording all API calls and related events (who, what, when, and from where) made via the AWS Management Console, SDKs, or other services.
    - **CloudTrail Logs**: Log files are delivered to an Amazon S3 bucket for secure, long-term storage.
    - **CloudTrail Insights**: Automatically detects unusual API activity or anomalies from your typical usage patterns.

Additional AWS Tools

- **AWS X-Ray**: Helps developers analyze and debug distributed applications, such as microservices and serverless architectures, by providing an end-to-end view of requests as they travel through different components.
- **AWS Config**: A configuration auditing tool that tracks AWS resource configurations and changes over time, helping to assess compliance with internal guidelines and industry regulations.

- **Amazon GuardDuty**: A threat detection service that continuously monitors for malicious activity and unauthorized behavior to protect your AWS accounts and workloads.
- **Amazon VPC Flow Logs**: Captures information about the IP traffic going to and from network interfaces in your VPC, useful for security analysis and troubleshooting network connectivity issues.

Third-Party and Open-Source Options

For multi-cloud or hybrid environments, or for more advanced visualizations, many third-party and open-source tools integrate with AWS, including Datadog, New Relic, Splunk Observability, Grafana, and Prometheus. These can offer unified observability across different platforms and potentially deeper application performance monitoring (APM) capabilities

5. Cost Management: Understanding billing models and optimizing cloud spend.

- Cost management is a major concern in cloud computing because cloud services typically follow a **pay-as-you-go** billing model. While this provides flexibility, it can also lead to unexpected expenses if resources are not managed carefully.
- Cloud cost management involves understanding pricing structures, tracking usage, setting budgets, and optimizing resource allocation. Common optimization strategies include rightsizing resources, shutting down unused services, and using reserved or discounted pricing options.

Billing Models (discuss the following in your own words with a scenario)

- Pay-as-you-go (On-Demand)
- Reserved Instances (RIs) / Savings Plans
- Spot Instances - unused
- Subscription/Fixed-Price
- Volume/Tiered Pricing

1. Pay-as-you-go (On-Demand)

Description:

Users pay only for the cloud resources they consume, without long-term commitments. Billing is typically per second, minute, or hour.

Key Features:

- No upfront payment, High flexibility
- Ideal for short-term or unpredictable workloads

Example Scenario:

A startup launches a new application and experiences uncertain traffic. Using on-demand pricing, the company pays only for the compute resources used during peak and off-peak hours, avoiding unnecessary costs.

2. Reserved Instances (RIs) / Savings Plans

Description:

Users commit to using certain cloud resources for a fixed period (usually 1 or 3 years) in exchange for significant discounts.

Key Features:

- Lower cost compared to on-demand, Requires long-term commitment
- Best for steady, predictable workloads

Example Scenario:

A company runs a database server 24/7. By purchasing a reserved instance, it reduces compute costs by up to 60% compared to on-demand pricing.

3. Spot Instances (Unused Capacity)

Description:

Spot instances use unused cloud provider capacity and are offered at very low prices. However, they can be interrupted when the provider needs the capacity back.

Key Features:

- Extremely cost-effective, No availability guarantee
- Suitable for fault-tolerant workloads

Example Scenario:

A research team runs batch data processing jobs. If a spot instance is interrupted, the job can restart without major impact, making it a cost-efficient choice.

4. Subscription / Fixed-Price Model

Description:

Users pay a fixed fee (monthly or annually) for access to a predefined set of cloud services or resources.

Key Features:

- Predictable billing, Simplified budgeting, Limited flexibility

Example Scenario:

A small business subscribes to a cloud-based email and collaboration platform with a fixed monthly fee per user, regardless of usage.

5. Volume / Tiered Pricing

Description:

Pricing decreases as usage increases. Higher consumption moves users into lower-cost tiers.

Key Features:

- Discounts for large-scale usage

- Encourages scaling, Common for storage and data transfer services

Example Scenario:

A video streaming service stores petabytes of data in cloud storage. As storage usage grows, the cost per gigabyte decreases, reducing overall expenses.

Cost Management Strategies & Features (discuss the following in your own words with a scenario)

1. Visibility & Monitoring: Track usage and spending
2. Transparency: Clear cost breakdown
3. Budgeting: Set and control spending limits
4. Right-Sizing: Match resources to needs
5. Resource Optimization: Remove waste
6. Tagging & Allocation: Track costs accurately
7. Leverage Discounts: Use pricing benefits
8. Automation: Enforce cost control automatically

Effective cloud billing management helps organizations control costs, improve accountability, and maximize the value of cloud investments.

1. Visibility & Monitoring

Description:

Visibility and monitoring involve tracking cloud usage and spending in real time using dashboards, reports, and alerts.

Key Benefits:

- Identifies cost spikes early, Tracks usage across teams and services

Example Scenario:

A company notices an unexpected increase in monthly cloud costs. Monitoring tools reveal that a test environment was left running 24/7, allowing the team to take corrective action.

2. Transparency

Description:

Transparency ensures that cloud costs are clearly broken down by service, department, or project.

Key Benefits:

- Improves accountability, Helps stakeholders understand spending patterns

Example Scenario:

Finance teams can see exactly how much each department spends on compute, storage, and networking, preventing billing disputes.

3. Budgeting

Description:

Budgeting allows organizations to set spending limits and receive alerts when costs approach or exceed defined thresholds.

Key Benefits:

- Prevents overspending, Supports financial planning

Example Scenario:

A startup sets a monthly cloud budget. When usage reaches 80% of the limit, an alert notifies the team to optimize resources.

4. Right-Sizing

Description:

Right-sizing involves matching resource capacity (CPU, memory, storage) to actual workload needs.

Key Benefits:

- Eliminates overprovisioning, Reduces unnecessary costs

Example Scenario:

An application server using only 20% CPU is downgraded to a smaller instance type, cutting costs without impacting performance.

5. Resource Optimization

Description:

Resource optimization focuses on improving efficiency by removing idle, unused, or redundant resources.

Key Benefits:

- Maximizes resource utilization, Lowers operational expenses

Example Scenario:

Unused storage volumes and unattached IP addresses are identified and deleted to reduce recurring charges.

6. Tagging & Allocation

Description:

Tagging assigns labels (such as project name, owner, or environment) to cloud resources to track and allocate costs.

Key Benefits:

- Enables cost allocation by team or project, Improves governance

Example Scenario:

Resources are tagged as “Development,” “Testing,” or “Production,” allowing cost reports to show spending per environment.

7. Leverage Discounts

Description:

Cloud providers offer discounts through reserved instances, savings plans, spot pricing, and volume discounts.

Key Benefits:

- Reduces long-term costs, Rewards predictable usage

Example Scenario:

A company running steady workloads purchases reserved instances, reducing compute costs significantly compared to on-demand pricing.

8. Automation

Description:

Automation uses scripts and policies to control cloud spending automatically.

Key Benefits:

- Reduces human error, Ensures consistent cost control

Example Scenario:

Non-production environments are automatically shut down after business hours, saving costs without manual intervention.

Activity: Explore the dashboard of AWS Billing & Cost Management service

Example Scenario:

A startup launches a cloud-based application and initially provisions large virtual machines to handle future growth. Over time, cost reports show that these machines are underutilized. By analysing usage data, the team downsizes the instances and schedules non-production environments to shut down after business hours, reducing monthly cloud costs significantly.

Effective cost management ensures that organizations gain the benefits of scalability and performance without overspending.